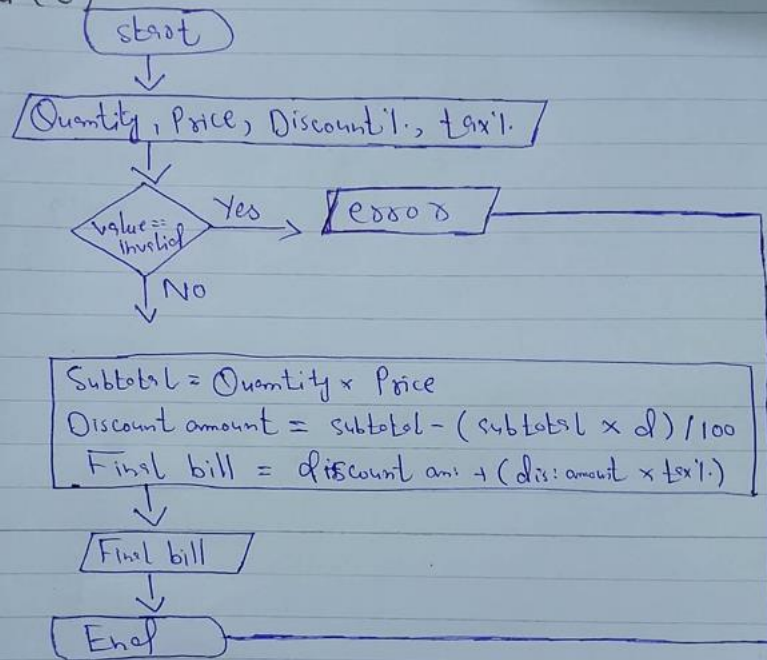


Q #14 (b)



Pseudocode

Start

input quantity, price, discount %, tax %.

if $q \leq 0$ or $p \leq 0$ or $d < 0$ or $d > 100$ or $t < 0$ then
Print "invalid input"
End

End if

$S = q \times p$

discounted $\leftarrow S - (S \times d) / 100$

Final bill $\leftarrow$ discounted + (discounted $\times t$) / 100

Print S, discounted, final bill

Generate bill document

End

PAC-Chart:

Input Program	Process	Output
	take input, validates, call calculation fun, Generate bills and display final amount	- Recd input - validate data - Call calculate subtotal → Cal: discount → Cal final bill - Generate bill document
Calculation	Performs 3 stage cal: using formula	$S = Q \times P$ $dis = S - (S \times d) / 100$ $F.B = dis + (dis \times t) / 100$
Validation module	Check for invalid values	- Check q, p, d, t - Show error message - terminate if invalid

IPO - Chart

Input	Process	Output
- Quant - Price - Discount - Tax	- validate input - Cal subtotal - Cal disc amount - Cal final bill - Generate bill document	- Subtotal - Discount amount - Final bill - Bill document

